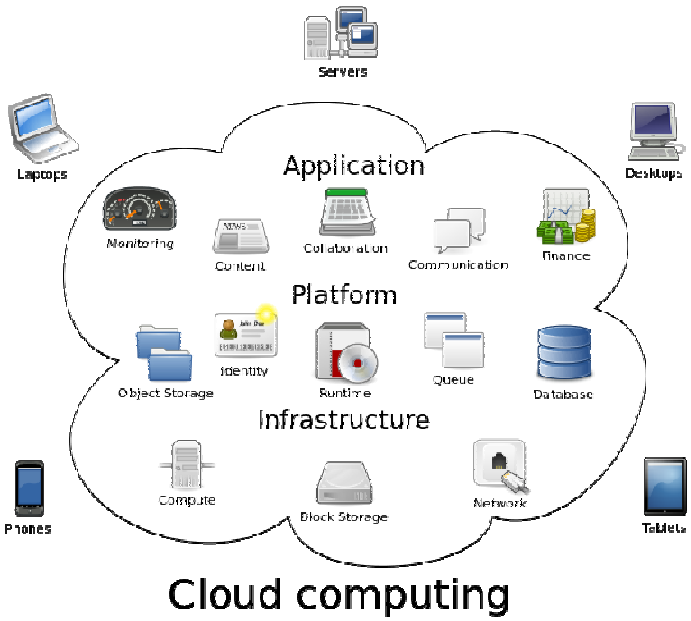
Defining the Cloud

The term cloud has been used as a metaphor for the Internet. This usage was originally derived from its common depiction in network diagrams as an outline of a cloud, used to represent the transport of data across carrier backbones (which owned the cloud) to an endpoint location on the other side of the cloud.



The cloud is a set of hardware, networks, storage, services, and interfaces that enable the delivery of computing as a service. Cloud services include the delivery of software, infrastructure, and storage over the internet (either as separate components or a complete platform) based on user demand.

Cloud computing is a pay-per-use model for enabling available, convenient, on-demand network access to a shared pool or configurable computing resources that can be rapidly provisioned and released with minimal management effort or service provider interaction.

Key characteristics of Cloud Computing:

- **On-demand self-service**

  A consumer can independently provision computing capabilities such as server time and network storage, as needed without requiring human interaction with each service's provider.

- **Ubiquitous(easily available from everywhere) network access**

  Capabilities are available over the network and accessed through standard mechanisms that promote use by heterogeneous thin or thick client platforms including mobiles, laptops etc.

- **Location-independent resource pooling**

  The consumer generally has no control over or knowledge of the exact location of the provided resources like storage, processing, memory, bandwidth etc.

- **Rapid elasticity**

  Capabilities can be rapidly and elastically provisioned to quickly scale up, and rapidly released to quickly scale down. To the consumer, the capabilities available for rent often appear to be infinite and can be purchased in any quantity at any time.

- **Pay-per-use**

  Capabilities are charged using a metered, fee-for-service, or advertising based billing model to promote optimization of resource use.

Basic building blocks of Cloud:

- **The end user**
  The end user doesn't really have to know anything about the underlying technology. In small businesses, for example, the cloud provider becomes the de facto data center. In larger organizations, the IT organization oversees the inner workings of both internal resources and external cloud resources.
- **Business management**
  Business management needs to take responsibility for overall governance of data or services living in a cloud. Cloud service providers must provide a predictable and guaranteed service level and security to all their constituents.
- **The cloud service provider**
  The cloud service provider is responsible for IT assets and maintenance.

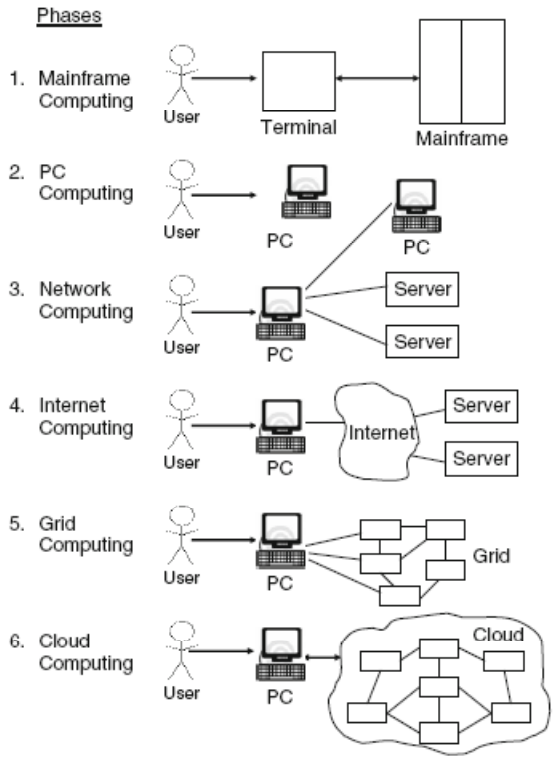
The Emergence of Cloud Computing

The underlying concept of cloud computing dates back to the 1950s; when large-scale mainframe became available in academia and corporations, accessible via thin clients /terminal computers.

In the 1990s, telecommunications companies, who previously offered primarily dedicated point-to-point data circuits, began offering virtual private network (VPN) services with comparable quality of service but at a much lower cost. By switching traffic to balance utilization as they saw fit, they were able to utilize their overall network bandwidth more effectively.

One of the first milestones for cloud computing was the arrival of Salesforce.com in 1999, which pioneered the concept of delivering enterprise applications via a simple website. The services firm paved the way for both specialist and mainstream software firms to deliver applications over the internet.

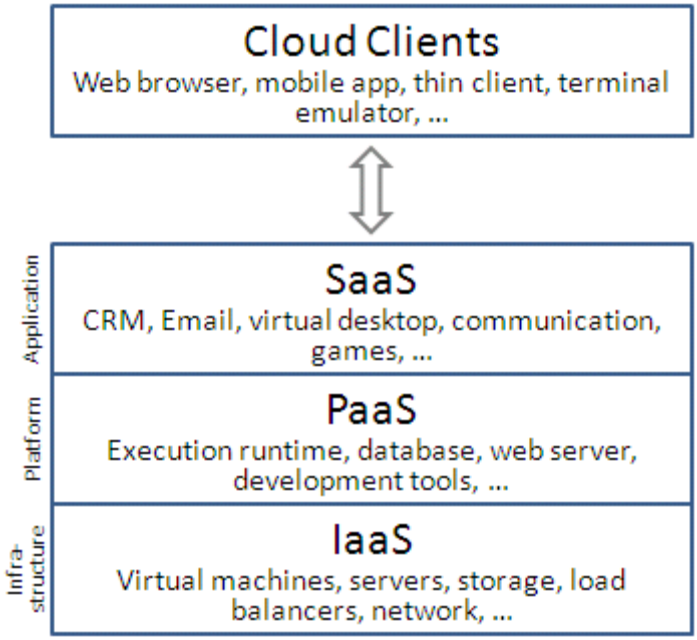
The next development was Amazon Web Services in 2002, which provided a suite of cloud-based services including storage, computation and even human intelligence through the Amazon Mechanical Turk. Then in 2006, Amazon launched its Elastic Compute cloud (EC2) as a commercial web service that allows small companies and individuals to rent computers on which to run their own computer applications.



Azure was announced in October 2008 and released on 1 February 2010 as Windows Azure, before being renamed to Microsoft Azure on 25 March 2014. It provides both PaaS and IaaS services and supports many different programming languages, tools and frameworks, including both Microsoft-specific and third-party software and systems.

Cloud-Based Services

The services given by the service provider to the customers or users through cloud computing technology are said cloud services. Service Provider's server gives both the hardware and software necessary and thus easy in management for both user and the cloud service provider. Cloud computing providers offer their services according to three fundamental models: Infrastructure as a service (IaaS), Platform as a service (PaaS), and Software as a service (SaaS) where IaaS is the most basic and each higher model abstracts from the details of the lower models.



a. Infrastructure as a service (IaaS)

IaaS refers not to a machine that does all the work, but simply to a facility given to businesses that offers users the leverage of extra storage space in servers and data centers. In this most basic cloud service model, providers offer computers, as physical or more often as virtual machines, and other resources. . Other resources in IaaS clouds include images in a virtual machine image library, raw (block) and file-based storage, firewalls, load balancers, IP addresses, virtual local area networks (VLANs), and software bundles. Examples of IaaS include: Amazon CloudFormation (and underlying services such as Amazon EC2), Rackspace Cloud, Terremark, Windows Azure Virtual Machines, Google Compute Engine, and Joyent.

b. Platform as a service (PaaS)

In the PaaS model, cloud providers deliver a computing platform typically including operating system, programming language execution environment, database, and web server. Application developers can develop and run their software solutions on a cloud platform without the cost and complexity of buying and managing the underlying hardware and software layers. Examples of PaaS include: Amazon Elastic Beanstalk, Cloud Foundry, Heroku, Force.com, EngineYard, Mendix, Google App Engine, Windows Azure Compute and OrangeScape.

c. Software as a service (SaaS)

In this model, cloud providers install and operate application software in the cloud and cloud users access the software from cloud clients. The cloud users do not manage the cloud infrastructure and platform on which the application is running. This eliminates the need to install and run the application on the cloud user's own computers simplifying maintenance and support. The pricing model for SaaS applications is typically a monthly or yearly flat fee per user, so price is scalable and adjustable if users are added or removed at any point. Examples of SaaS include: Google Apps, Microsoft Office 365, and Onlive.

Grid Computing or Cloud Computing?

Grid computing is often confused with cloud computing.

Grid computing is where more than one computer coordinates to solve a problem together. Often used for problems involving a lot of number crunching, which can be easily parallelisable.

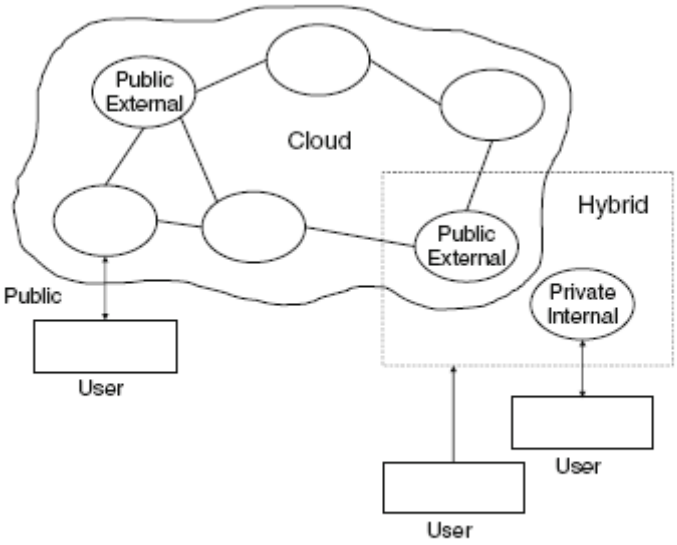
Cloud computing is where an application doesn't access resources it requires directly, rather it accesses them through something like a service. So instead of talking to a specific hard drive for storage, and a specific CPU for computation, etc. it talks to some service that provides these resources. The service then maps any requests for resources to its physical resources, in order to provide for the application. Usually the service has access to a large amount of physical resources, and can dynamically allocate them as they are needed.

A cloud would usually use a grid. A grid is not necessarily a cloud or part of a cloud.

Several deployment models for cloud computing

- **Private cloud**

  The infrastructure is operated solely for an organization; it may be managed by the organization or a third party and may exist on or off the premises of the organization. It is also referred to as 'corporate' or 'internal' Cloud, term used to denote a proprietary computing architecture providing hosted services on private networks.



- **Public cloud**

  The infrastructure is made available to the general public or a large industry group and is owned by an organization selling cloud services. It is also referred to as 'external' Cloud that describes the conventional meaning of Cloud computing: scalable, dynamically provisioned, often virtualized resources available over the internet from an off-site third party provider, which divides up resources and bills its customers on a 'utility' basis.

- **Community cloud**

  Community cloud shares infrastructure between several organizations from a specific community with common concerns (security, compliance, jurisdiction, etc.), whether managed internally or by a third-party and hosted internally or externally. The costs are spread over fewer users than a public cloud (but more than a private cloud), so only some of the cost savings potential of cloud computing are realized.

- **Hybrid cloud**

  Here, the infrastructure is a composition of two or more clouds (private, or public) that remain unique entities but are bound together by standardized or proprietary technology that enables data and application portability (e.g., cloud bursting for load-balancing between clouds). Cloud bursting is an application deployment model in which an application runs in a private cloud or data center and bursts into a public cloud when the demand for computing capacity spikes. The advantage of such a hybrid cloud deployment is that an organization only pays for extra compute resources when they are needed.

Benefits of using cloud models

- **Cost Savings**

  Companies can reduce their capital expenditures and use operational expenditures for increasing their computing capabilities. This is a lower barrier to entry and also requires fewer in-house IT resources to provide system support.

- **Scalability/Flexibility**

  Companies can start with a small deployment and grow to a large deployment fairly rapidly, and then scale back if necessary. Also, the flexibility of cloud computing allows companies to use extra resources at peak times, enabling them to satisfy consumer demands.

- **Reliability**

  Services using multiple redundant sites can support business continuity and disaster recovery.

- **Maintenance**

  Cloud service providers do the system maintenance, and access is through APIs that do not require application installations onto PCs, thus further reducing maintenance requirements.

- **Mobile Accessible**

  Mobile workers have increased productivity due to systems accessible in an infrastructure available from anywhere.

Characteristics of Cloud Computing

Cloud computing has a variety of characteristics, with the main ones being:

- **Shared Infrastructure**

  Uses a virtualized software model, enabling the sharing of physical services, storage, and networking capabilities. The cloud infrastructure, regardless of deployment model, seeks to make the most of the available infrastructure across a number of users.

- **Dynamic Provisioning / on demand self service**

  Allows for the provision of services based on current demand requirements. This is done automatically using software automation, enabling the expansion and contraction of service capability, as needed. This dynamic scaling needs to be done while maintaining high levels of reliability and security.

- **Broad Network Access**

  Needs to be accessed across the internet from a broad range of devices such as PCs, laptops, and mobile devices, using standards-based APIs (for example, ones based on HTTP). Deployments of services in the cloud include everything from using business applications to the latest application on the newest smartphones.

- **Multi-Tenant Capable**

  The resources (e.g., network, storage and compute power) can be shared among multiple enterprise clients, thereby lowering overall expense. Resource virtualization is used to enforce isolation and aid in security.

- **Rapid Elasticity**

  The consumer should have the ability to rapidly (often automatically) increase or decrease the computing resources needed to carry out their work.

- **Managed Metering / Measured Service**

  Uses metering for managing and optimizing the service and to provide reporting and billing information. In this way, consumers are billed for services according to how much they have actually used during the billing period.

Challenges for cloud computing

- **Security and Privacy**

  These issues are generally attributed to slowing the deployment of cloud services. These challenges can be addressed, for example, by storing the information internal to the organization, but allowing it to be used in the cloud. For this to occur, though, the security mechanisms between organization and the cloud need to be robust and a Hybrid cloud could support such a deployment.
- **Lack of Standards**

  Clouds have documented interfaces; however, no standards are associated with these, and thus it is unlikely that most clouds will be interoperable.
- **Continuously Evolving**

  User requirements are continuously evolving, as are the requirements for interfaces, networking, and storage. This means that a "cloud," especially a public one, does not remain static and is also continuously evolving.
- **Compliance Concerns**

  The EU has a legislative backing for data protection across all member states, but in the US data protection is different and can vary from state to state. As with security and privacy mentioned previously, these typically result in Hybrid cloud deployment with one cloud storing the data internal to the organization.

Grid Computing

Grid computing is a processor architecture that combines computer resources from various domains to reach a main objective. In grid computing, the computers on the network can work on a task together, thus functioning as a supercomputer.

Typically, a grid works on various tasks within a network, but it is also capable of working on specialized applications. It is designed to solve problems that are too big for a supercomputer while maintaining the flexibility to process numerous smaller problems. Computing grids deliver a multiuser infrastructure that accommodates the discontinuous demands of large information processing.

The idea of grid computing was first established in the early 1990s by Carl Kesselman, Ian Foster and Steve Tuecke. They developed the Globus Toolkit standard, which included grids for data storage management, data processing and intensive computation management.

Grid computing is made up of applications used for computational computer problems that are connected in a parallel networking environment. It connects each PC and combines information to form one application that is computation-intensive.

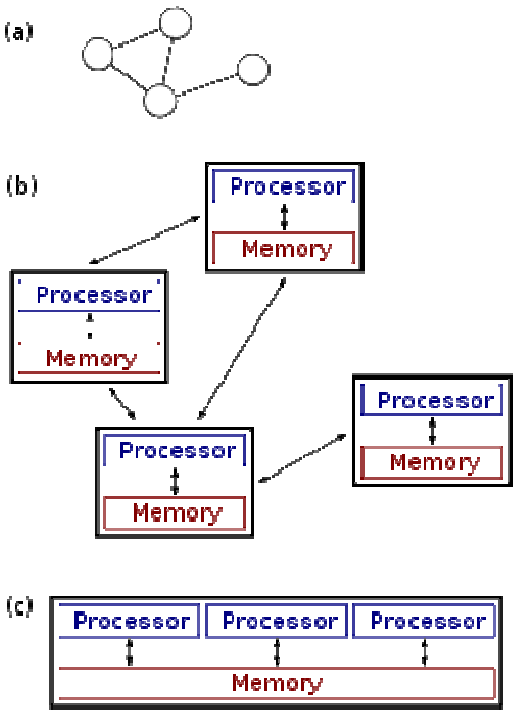
Distributed Computing in Grid and Cloud

Distributed computing is a field of computer science that studies distributed systems. A distributed system consists of multiple autonomous computers that communicate through a computer network. The computers interact with each other in

order to achieve a common goal. In distributed computing, a problem is divided into many tasks, each of which is solved by one or more computers, which communicate with each other by message passing. While there is no single definition of a distributed system, the following defining properties are commonly used:

- There are several autonomous computational entities, each of which has its own local memory.
- The entities communicate with each other by message passing

Figure (a) is a schematic view of a typical distributed system; as usual, the system is represented as a network topology in which each node is a computer and each line connecting the nodes is a communication link. Figure (b) shows the same distributed system in more detail: each computer has its own local memory, and information can be exchanged only by passing messages from one node to another by using the available communication links. Figure (c) shows a parallel system in which each processor has a direct access to a shared memory.



Communication-as-a-Service (CaaS)

CaaS is an outsourced enterprise communications solution that can be leased from a single vendor. Such communications can include voice over IP (VoIP or Internet telephony), instant messaging (IM), collaboration and videoconference applications using fixed and mobile devices. CaaS brings social networking, cloud computing, and smartphones together, providing cloud- technologies that let users communicate via voice, text, and rich media on whatever device they prefer to use.

The CaaS vendor is responsible for all hardware and software management and offers guaranteed Quality of Service (QoS). CaaS allows businesses to selectively deploy communications devices and modes on a pay-as-you-go, as-needed basis. This approach eliminates the large capital investment and ongoing overhead for a system whose capacity may often exceed or fall short of current demand.

CaaS offers flexibility and expandability that small and medium-sized business might not otherwise afford, allowing for the addition of devices, modes or coverage on demand. The network capacity and feature set can be changed from day to day if necessary so that functionality keeps pace with demand and resources are not wasted. There is no risk of the system becoming obsolete and requiring periodic major upgrades or replacement.

Advantages of CaaS

- Fully Integrated Enterprise Class Unified Communication
  By managing the LAN/WAN, the vendor can guarantee consistent Quality of Service (QoS) from the desktop across the VoIP backbone and back again. Advanced Unified Communications features such as Outlook integration, soft phones, real-time presence, chat, multimedia conferencing, video calling, unified messaging and mobility are also part of a standard CaaS deployment.
- No Upfront Capital Expenses
  Since cloud services are supposed to lower capital expenditure and focus more on operating expenditure, by implementing CaaS, consumers can build up their communication infrastructure without any upfront cost. They just need to pay it as a service.
- Flexibility in Features
  Since cloud is a multi-tenant architecture, cloud vendors have to manage multiple customers and look after the features that they want. Economies of scale also mean that the service provider is not tied to a single vendor investment and can leverage best- of-breed providers like Cisco, Microsoft and Nortel much more economically than an independent enterprise.
- No Risk of Obsolescence
  Technology changes rapidly and are obsolete within few years of introduction. With CaaS, companies are always privileged with new technologies as cloud vendors keep on updating their equipment's and technologies to sustain in market.
- No Data Center Cost
  As a prime advantages of cloud computing, while using CaaS infrastructure, organization need not invest on expensive servers, cooling system and electric equipment. With monthly/ yearly recurring cost, organization can dramatically cut down the management cost of data center as well.
- Guaranteed Business Continuity

With CaaS, organization can be hugely benefitted with guaranteed business continuity as cloud service providers proactively plans for Business Continuity Planning for their customers. Service uptime is guaranteed even if any catastrophic disaster strikes.

Enterprise-Class Unified Communications

Unified Communications (UC) might be defined as communications integrated to optimize business performance, and can describe a seamless set of voice, video and Web collaboration applications designed to enable advanced connectivity between employees, customers, partners and other stakeholders available on any device.

In the large enterprise setting, Unified Communications allow employees to send and receive messages on various medium – guided by presence information or employee preference – switching calls between home lines and a mobile device, or accessing a voicemail via text or email.

When well implemented, UC can save money, optimize business processes and improve communications by better connecting devices, media and applications, reducing latency, and opening new forms of collaboration.

Benefits of Unified Communication

- Efficient Communication
  Users can communicate more efficiently by having access to all communications at one time and being free to share, forward, or manage them in the way that's most convenient or effective for the given communication.
- Anywhere Access
  Unified messaging provides alternative methods of accessing communications. By merging e-mail, voice, and other communications, users can get voice messages in e-mail, have e-mail dictated over the phone, or access communications via the Web.
- Collaboration
  Collaborative approach of communication can be helpful in achieving immediacy, simplicity and interoperability of communication. In this way, business can sense and respond more quickly to changing environment.
- Business Process Integration
  Business processes can be integrated to each other for more efficiency. Many business entities and functions can be incorporated in one platform to make processes more agile and responsive.

Infrastructure-as-a-Service (IaaS)

It is a provision model in which an organization outsources the equipment used to support operations, including storage, hardware, servers and networking components. The service provider owns the equipment and is responsible for housing, running and maintaining it. The client typically pays on a per-use basis.

Clients are able to self-provision this infrastructure, using a Web-based graphical user interface that serves as an IT operations management console for the overall environment.

In IaaS, customers rent computing resources rather than purchase them and access the resources through a (Web) service infrastructure. Service billed typically monthly on a usage basis. Example: Amazon EC2

Advantages

- Access to preconfigured environment
- Use of latest technology
- Reduced cost and risk of having third party maintain resources
- No capital investment
- No IT personal to maintain remote hardware/software
- Able to manage peak demand as needed without having to purchase a larger system that would be underutilized at other times
- Secure – security handed by provider

Disadvantages

- Delays in network (Internet)
- Confidential data concerns

Features and Benefits of IaaS

- Scalability
  Resource is available as and when the client needs it and, therefore, there are no delays in expanding capacity or the wastage of unused capacity
- No investment in hardware
  The underlying physical hardware that supports an IaaS service is set up and maintained by the cloud provider, saving the time and cost of doing so on the client side
- Utility style costing
  The service can be accessed on demand and the client only pays for the resource that they actually use
- Location independence
  The service can usually be accessed from any location as long as there is an internet connection and the security protocol of the cloud allows it
- Physical security of data center locations
  Services available through a public cloud, or private clouds hosted externally with the cloud provider, benefit from the physical security afforded to the servers which are hosted within a data center
- No single point of failure
  If one server or network switch, for example, were to fail, the broader service would be unaffected due to the remaining multitude of hardware resources and redundancy configurations. For many services if one entire data center were to go offline, never mind one server, the IaaS service could still run successfully.

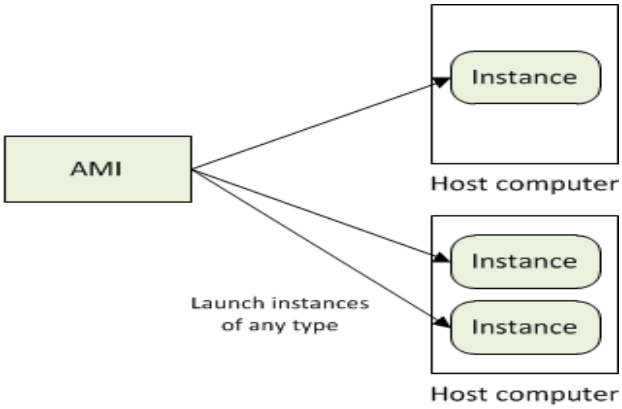
IaaS: On-Demand Computing

On-demand (OD) computing is an increasingly popular enterprise model in which computing resources are made available to the user as needed. The resources may be maintained within the user's enterprise, or made available by a service provider. The on-demand model evolved to overcome the challenge of being able to meet fluctuating resource demands efficiently. Because demand for computing resources can vary drastically from one time to another, maintaining sufficient resources to meet peak requirements can be costly.

On-demand computing products are rapidly becoming prevalent in the marketplace. Computer Associates, HP, IBM, Microsoft, and Sun Microsystems are among the more prominent on-demand vendors. These companies refer to their on-demand products and services by a variety of names. Concepts such as grid computing, utility computing, autonomic computing, and adaptive management seem very similar to the concept of on-demand computing. The major advantage of On Demand Computing (ODC) is low initial cost, as computational resources are essentially rented when they are required. This provides cost savings over purchasing them outright.

IaaS: Amazon Elastic Cloud (EC2)

Amazon Elastic Compute Cloud (EC2) is a central part of Amazon cloud computing platform Amazon Web Services (AWS). EC2 allows users to rent virtual computers on which to run their own computer applications. EC2 allows scalable deployment of applications by providing a Web service through which a user can boot an Amazon Machine Image to create a virtual machine, which Amazon calls an "instance", containing any software desired.



A user can create, launch, and terminate server instances as needed, paying by the hour for active servers, hence the term "elastic". It reduces the time required to obtain and boot new server instances to minutes, allowing customers to quickly scale capacity as their computing demands dictate.

It changes the economics of computing by allowing clients to pay only for capacity they actually use. It provides developers the tools needed to build failure-resilient applications and isolate themselves from common failure scenarios.

To use the EC2, a subscriber creates an Amazon Machine Image (AMI) containing the operating system, application programs and configuration settings. Then the AMI is uploaded to the Amazon Simple Storage Service (Amazon S3) and registered with Amazon EC2, creating a so-called AMI identifier (AMI ID). Once this has been done, the subscriber can requisition virtual machines on an as-needed basis. Capacity can be increased or decreased in real time from as few as one to more than 1000 virtual machines simultaneously. Billing takes place according to the computing and network resources consumed.

The idea of EC2 is to lighten the cost of buying servers to host a system, but more importantly to eliminate the wasted time systems engineers devote to managing hard assets. Instead of buying servers to increase capacity or add new features, we simply buy more gigabytes on EC2. Amazon sells it in subscription form, with subscriptions based on how much capacity we use.

IaaS: Characteristics of Amazon EC2

- Persistent Storage
  Provide persistent storage independent of the lifetime of the EC2 instance, and act much like hard drives on a real server. More accurately, they appear as block devices to the operating system that are backed by Amazon's disk arrays.

- Elastic IP Addresses
  Amazon's elastic IP address feature is similar to static IP address in traditional data centers, with one key difference. A user can programmatically map an elastic IP address to any virtual machine instance without a network administrator's help and without having to wait for DNS to propagate the new binding. In this sense an Elastic IP Address belongs to the account and not to a virtual machine instance. It exists until it is explicitly removed, and remains associated with the account even while it is associated with no instance.
- Amazon Cloud Watch
  Amazon CloudWatch is a Web service that provides real-time monitoring to Amazon's EC2 customers on their resource utilization such as CPU, disk and network. Cloudwatch does not provide any memory, disk space, or load average metrics without running additional software on the instance. The metrics collected by Amazon CloudWatch enables Auto Scaling feature to dynamically add or remove EC2 instances. The customers are charged by the number of monitoring instances.
- Automated Scaling
  It allows to automatically adapt computing capacity to site traffic. The schedule-based (e.g. time-of-the-day) and rule-based (e.g. CPU utilization thresholds) auto scaling mechanisms are easy to use and efficient for simple applications.
- Reliability
  To make EC2 more fault-tolerant, Amazon engineered Availability Zones that are designed to be insulated from failures in other availability zones. High availability database services, like Amazon Relational Database Service run on top of EC2 instances.

Monitoring as a Service (MaaS)

Monitoring-as-a-Service is an outsourced service to provide security mainly to platforms that are run on the Internet for conducting business. Since the advent of Cloud Computing, the MaaS popularity has increased even more. Safe monitoring involves protecting a company or other institution / organization from cyber threats, in which a team prepared is crucial to maintain the confidentiality, integrity and access to IT assets.

Many industry regulations require organizations to monitor their security environment, server logs, and other information assets to ensure the integrity of these systems. However, conducting effective security monitoring can be a daunting task because it requires advanced technology, skilled security experts, and scalable processes—none of which come cheap. MaaS security monitoring services offer real-time, 24/7 monitoring and nearly immediate incident response across a security infrastructure - they help to protect critical information assets of their customers.

MaaS: Protection against Internal and External Threats

Security monitoring services can improve the effectiveness of a customer security infrastructure by actively analyzing logs and alerts from infrastructure devices around the clock and in real time. Typical services provided by many MaaS vendors are:

- Early Detection
  An early detection service detects and reports new security vulnerabilities shortly after they appear. Generally, the threats are correlated with third-party sources, and an alert or report is issued to customers. This report is usually sent by email to the person designated by the company. Security vulnerability reports, aside from containing a detailed description of the vulnerability and the platforms

affected, also include information on the impact the exploitation of this vulnerability would have on the systems or applications previously selected by the company receiving the report. Most often, the report also indicates specific actions to be taken to minimize the effect of the vulnerability, if that is known.

- Platform, Control, and Services Monitoring
  Platform, control, and services monitoring is often implemented as a dashboard interface and makes it possible to know the operational status of the platform being monitored at any time. It is accessible from a web interface, making remote access possible. Each operational element that is monitored usually provides an operational status indicator, always taking into account the critical impact of each element. This service aids in determining which elements may be operating at or near capacity or beyond the limits of established parameters. By detecting and identifying such problems, preventive measures can be taken to prevent loss of service.
- Intelligent Log Centralization and Analysis
  Intelligent log centralization and analysis is a monitoring solution based mainly on the correlation and matching of log entries. Such analysis helps to establish a baseline of operational performance and provides an index of security threat. Alarms can be raised in the event an incident moves the established baseline parameters beyond a stipulated threshold. These types of sophisticated tools are used by a team of security experts who are responsible for incident response once such a threshold has been crossed and the threat has generated an alarm or warning picked up by security analysts monitoring the systems.
- Vulnerabilities Detection and Management
  Vulnerabilities detection and management enables automated verification and management of the security level of information systems. The service periodically performs a series of automated tests for the purpose of identifying system weaknesses that may be exposed over the Internet, including the possibility of unauthorized access to administrative services, the existence of services that have not been updated, the detection of vulnerabilities such as phishing, etc. The service performs periodic follow-up of tasks performed by security professionals managing information systems security and provides reports that can be used to implement a plan for continuous improvement of the systems security level.
- Intervention, Forensics, and Help Desk Services
  Quick intervention when a threat is detected is crucial to mitigating the effects of a threat. This requires security engineers with ample knowledge in the various technologies and with the ability to support applications as well as infrastructures on a 24/7 basis. MaaS platforms routinely provide this service to their customers. When a detected threat is analyzed, it often requires forensic analysis to determine what it is, how much effort it will take to fix the problem, and what effects are likely to be seen. When problems are encountered, the first thing customers tend to do is pick up the phone. This service includes assistance in writing failure reports, managing operating problems, etc.

Platform as a Service (PaaS)

Platform as a service (PaaS) is a category of cloud computing services that provide a computing platform and a solution stack as a service. PaaS offerings facilitate the deployment of applications without the cost and complexity of buying and managing the underlying hardware and software and provisioning hosting capabilities. Technically, a PaaS is an Application Platform comprised of an operating system, middleware and other software that allows applications to run on the cloud with much of the management, security, scaling and other stack related headaches abstracted away. This allows us to focus on two things: customers and developing your application. Platform as a Service allows users to create software applications using tools supplied by the provider.

PaaS services can consist of preconfigured features that customers can subscribe to; they can choose to include the features that meet their requirements while discarding those that do not. PaaS works on top of IaaS and will do all of that work automatically.

PaaS: The Traditional On-Premises Model

The traditional approach to developing and running applications on-premises has always been complex, expensive and risky. Producing our own solution never brought any guarantee of success. Each application has been designed to meet their specific requirements within each business. Each solution requires a specific programming hardware, an operating system, a database, often a middle-ware package, mail and web servers, etc. Once environment was created in hardware and software, a team of developers had to navigate a complex programming platform to build their own applications. Additionally, a team of network, database and system management was needed to keep everything in perfect driving conditions. Inevitably, developers were forced to change the application on behalf of a detail of the business, generating new cycles of testing before being distributed.

PaaS model offers a choice of faster and more cost-effective application development and delivery. Furthermore, it provides all the infrastructure needed to run applications over the Internet. Just like Google, iTunes and Youtube, this cloud computing model allows new functionality to be delivered in emerging markets through web browsers. PaaS is based on a model of mediation or subscription, and users only pay for what they use. The PaaS model in its range also includes other facilities such as, application design and development, testing, deployment and hosting as well as integration, security, scalability, storage, status management, control panel, etc.

PaaS:Characteristics

- Multi-tenant architecture
  A PaaS offering must be multi-tenanted where one that uses common computing resources including hardware, operating system, software (i.e. application code), and a single underlying database with a shared schema to support multiple customers simultaneously. It has a direct contrast to the traditional client/server architecture, which requires an entire stack of hardware and software to be dedicated to each tenant (customer).
- Customizable /Programmable User Interface
  PaaS offering should provide the ability to construct highly flexible user interfaces via a simple “drag & drop” methodology that permits the creation and configuration of UI components on the fly. Furthermore, given the growing set of Web devices, additional flexibility to use other technologies such as CSS, AJAX etc to specify the appearance of the application’s interface should be available to the UI designer.
- Unlimited Database Customizations
  Database used by application should have option of customization for more flexibility in application development. Specifying relationships between objects, a key requirement of any sophisticated business application, must be possible through the declarative Web-based interface. Other mandatory functions include the ability to incorporate validation rules and permissions at the object/field level and the ability to specify auditing behavior.
- Automation
  PaaS environments automate the process of deploying applications to infrastructure, configuring application components, provisioning and configuring supporting technology like load balancers and databases, and managing system change based on policies set by the user.

While IaaS is known for its ability to shift capital costs to operational costs through outsourcing, only PaaS is able to cut down costs across the development, deployment and management aspects of the application lifecycle.

- Security
  The PaaS offering should provide a flexible access control system that allows detailed control over what users of the SaaS application can see and the data each user can access. Definition of access from the application level (including tabs, menus, objects, views, charts, reports and workflow actions) to the individual field level should be possible. Defining an access control model should be possible through the creation of groups and roles and the assignment of users to either groups or roles.
- Runtime Framework
  PaaS runtime framework executes end-user code according to policies set by the application owner and cloud provider. PaaS runtime frameworks come in many flavors, some based on traditional application runtimes, others based on 4GL and visual programming concepts, and some with pluggable support for multiple application runtimes.

Software as a Service (SaaS)

Software as a service , sometimes referred to as "on-demand software", is a software delivery model in which software and associated data are centrally hosted on the cloud. SaaS is typically accessed by users using a thin client via a web browser. In this model, the software is not hosted on the customers' individual computers. Under the SaaS model, a vendor is responsible for the creation, updating, and maintenance of software. Customers buy a subscription to access it, which includes a separate license, or seat, for each person that will use the software.

SaaS has become a common delivery model for many business applications, including accounting, collaboration, customer relationship management (CRM), management information systems(MIS), enterprise resource planning (ERP), invoicing, human resource management (HRM), content management (CM) and service desk management. The emergence of SaaS as an effective software-delivery mechanism creates an opportunity for IT departments to change their focus from deploying and supporting applications to managing the services that those applications provide.

Unlike traditional software which is conventionally sold as a perpetual license with an up-front cost (and an optional ongoing support fee), SaaS providers generally price applications using a subscription fee, most commonly a monthly fee or an annual fee. Consequently, the initial setup cost for SaaS is typically lower than the equivalent enterprise software. SaaS vendors typically price their applications based on some usage parameters, such as the number of users ("seats") using the application. However, because in a SaaS environment customers' data reside with the SaaS vendor, opportunities also exist to charge per transaction, event, or other unit of value.

Saas: Characteristics

- Simple and quick system implementation
  As Saas sits on top of PaaS and IaaS, deploying any enterprise level software becomes easy and quick as SaaS inherits all the features of underlying infrastructure. Also since SaaS is scalable, any user request can be addressed with required elasticity.
- Transparent and low pricing

With common infrastructure, cloud vendor can leverage their infrastructure with lower cost and transparency. End result of this directly impacts customer cost of software implementation as they gets it cheaper with enhanced security and high availability.

- Multitenant Architecture
  A multitenant architecture, in which all users and applications share a single, common infrastructure and code base that is centrally maintained. Because SaaS vendor clients are all on the same infrastructure and code base, vendors can innovate more quickly and save the valuable development time previously spent on maintaining numerous versions of outdated code.
- Easy software maintenance and Customization
  The ability for each user to easily customize applications to fit their business processes without affecting the common infrastructure. Because of the way SaaS is architected, these customizations are unique to each company or user and are always preserved through upgrades. That means SaaS providers can make upgrades more often, with less customer risk and much lower adoption cost.
- Better Access
  Improved access to data from any networked device while making it easier to manage privileges, monitor data use, and ensure everyone sees the same information at the same time.

SaaS: Benefits

- Save money by not having to purchase servers or other software to support use.
- Focus Budgets on competitive advantage rather than infrastructure.
- Monthly obligation rather than up front capital cost.
- Reduced need to predict scale of demand and infrastructure investment up front as available capacity matches demand.
- Multi-Tenant efficiency
- Flexibility and scalability
- Security

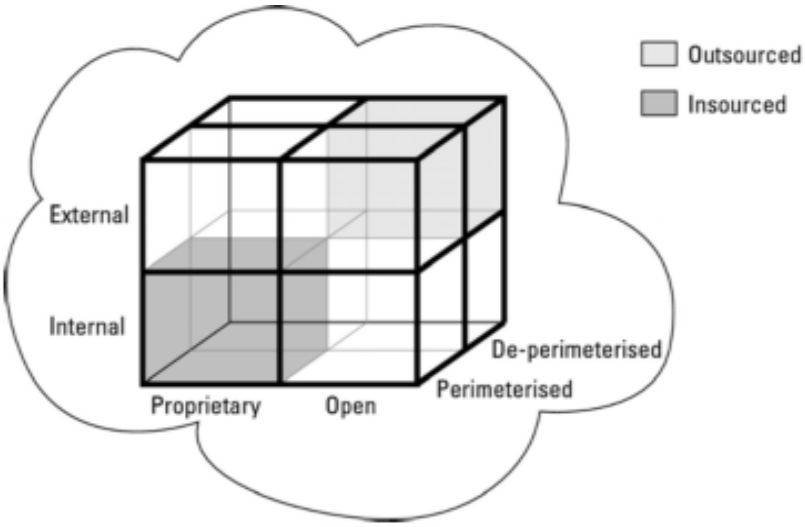
SaaS: Implementation Issues

- **An Internet connection is required at all times**
  One of the biggest drawbacks of SaaS is the fact that employee's can no longer work offline when SaaS software services are used and that they must be connected to the internet whenever they need to use these SaaS software services. For employees working from within their businesses headquarters or for employees who have a dedicated internet connection at home, this is not too much of a problem. However, for employees who work from a mobile device such as a laptop or for employees who are constantly on the move, this is obviously a problem.
- **The security and control of our data is no longer in our own hands**
  When using SaaS, some if not all of the data of a business will be stored on their SaaS service provider's data centres. A business has no idea how their SaaS service provider will secure their data and what backup procedures that their service provider will have in place. The only thing that a business can go by, is the word of their service provider confirming that they will ensure that their clients data will be well looked after. To ensure that their data is well looked after, a business must ensure that their service provider writes down exactly how their clients data will be secured, in their Service Level Agreement contract.

- **We can no longer control what version of a software application that we are going to use**
  Although SaaS is great because software applications are constantly up to date, sometimes a business is better of using an older version of a software application but SaaS just does not allow this. For example businesses may wish to use an older version of a software application because this older version is stable or because it is compatible with their existing software applications but with SaaS, businesses only have access to the most current versions of an application. Again this is good because a business is constantly up to date but the fact that businesses do not have the choice to use older versions of a software application is the negative thing.
- **We lose our freedom and are effectively at the control of our software vendor**
  With a traditional software application, a business pays for this software application with a single upfront payment. After this, the software application is effectively in their control and they can do whatever they want with it (as long as they are not breaching their software vendor's terms and conditions, such as copyright, licensing, etc). However, with SaaS software services a business must continuously keep paying their SaaS service provider in order to access the software services that they require. If they accidentally fail to pay their service provider for one month (if they are paying monthly), then they risk losing their software services. If their SaaS service provider changes its payment rate, then a business has to pay for their software services at this revised payment rate. If a SaaS service provider faces internal problems, then their clients will also be affected by these problems.

Jericho Cloud Cube Model

The Open Group maintains an association called the Jericho Forum (https://www.opengroup.org/jericho/index.htm) whose main focus is how to protect cloud networks. The group has an interesting model that attempts to categorize a cloud network based on four dimensional factors.



The four dimensions of the Cloud Cube Model are shown in figure and listed here:

- **Physical location of the data**: Internal (I) / External (E) determines your organization's boundaries.
- **Ownership**: Proprietary (P) / Open (O) is a measure of not only the technology ownership, but of interoperability, ease of data transfer, and degree of vendor application lock-in.
- **Security boundary**: Parameterized (Per) / De-perimiterized (D-p) is a measure of whether the operation is inside or outside the security boundary or network firewall.
- **Sourcing**: Insourced or Outsourced means whether the service is provided by the customer or the service provider. Taken together, the fourth dimension corresponds to two different states in the eight possible cloud forms: Per (IP, IO, EP, EO) and D-p (IP, IO, EP, EO). The sourcing dimension addresses the deliverer of the service. What the Cloud Cube Model is meant to show is that the traditional notion of a network boundary being the network's firewall no longer applies in cloud computing.

Managed Service Providers

A managed service provider (MSP) is a third-party contractor that delivers network-based services, applications and equipment to enterprises, residences or other service providers. Managed service providers can be hosting companies or access providers that offer IT services such as fully outsourced network management arrangements, including IP telephony, messaging and call center management, virtual private networks (VPNs), managed firewalls and monitoring/reporting of network servers. Most of these services can be performed from outside a company's internal network with a special emphasis placed on integration and certification of Internet security for applications and content. MSPs serve a outsourcing agents for companies, especially other service providers like ISPs, that don't have the resources to constantly upgrade or maintain faster and faster computer networks.

Managed services providers can offer services such as alerts, security, patch management, data backup and recovery for different client devices: desktops, notebooks, servers, storage systems, networks and applications. Offloading routine infrastructure management to an experienced managed service professional lets us concentrate on running our business, with fewer interruptions due to IT issues. MSPs act as an extension of our IT department, taking care of routine IT infrastructure monitoring and management around the clock—freeing up our IT staff to focus on higher-value projects. By proactively monitoring and maintaining our systems, an MSP can help us avoid many technology problems in the first place. If an issue occurs, an experienced MSP can troubleshoot and resolve it more efficiently.

Evolution from Managed Service to Cloud Computing

Managed service providers are feeling a bit of pressure from cloud computing. After years of investing in infrastructure and business model changes to deliver remote managed services, MSPs are seeing the cloud as a threat to their livelihood. In many cases, the cloud requires no remote infrastructure or on- premise equipment. The fear is the cloud could render MSPs obsolete. Cloud computing could also be a managed service.

Cloud computing consists of security monitoring, storage management, network administration and the all managed services need not necessarily be cloud computing. Cloud computing puts its efforts in creating a technical solution. It is a technical model that delivers technical access to the computing resources. Thus, cloud computing can be defined as a technical solution. Managed services, on the other hand, is a contract based relationship. The definition and delivery of the service is done on a repeating revenue basis. It means that managed services are recurring revenues from well-defined services that are predictable.

Some types of services that managed service provider’s offer are help desk assistance and network administration services. Managed services allow the Information Technology staff of the company to focus their efforts and energy in to the core activities of the company rather than dealing with the IT challenges. A managed service is the management of technology like telephony, IT, applications and others. However, the definition of a managed service is changing.

There are, however a lot of similarities that exist between a managed service and cloud computing. While cloud computing is generally a technical implementation that decides how the infrastructure and applications are to be delivered over the private and public networks, it can also be a business model. The same contract agreement holds ground in a cloud computing relationship. Managed services can also be a technical implementation.

Cloud Optimized Infrastructure is based around 5 key belief that offer capabilities that differentiate as an MSP from cloud computing:

- Flexible licensing which supports the elastic expansion/contraction of cloud-based services and accommodates the billing implications
- Low overhead in deployment and use across server, network, and storage resources, making it a great fit for virtual machine environments that are a key supporting technology in cloud-based computing
- Non-disruptive scalability which accommodates the need for server, storage and other infrastructure growth on both the end user and cloud provider sides without impacting client-side production servers
- Enterprise multi tenancy that provides for the secure delivery of reliable services to multiple customers with a scalable management model
- Broad heterogeneous support that maximizes cloud provider market opportunities by covering a wide range of server, storage, and application environments found in customer settings

Single Purpose Architecture to Multipurpose Architecture

In the early days of MSPs, the providers would actually go onto customer sites and perform their services on customer-owned premises. Over time, these MSPs specialized in implementation of infrastructure and quickly figured out ways to build out data centers and sell those capabilities off in small chunks commonly known as monthly recurring services, in addition to the basic fees charged. With virtualization and other supporting technologies, MSP's quickly convinced their customers to shift their data centers to multipurpose and multitenant architecture. Not only multipurpose architecture was effective for their customers but also was huge cost saving initiative.

Data Center Virtualization

Data center virtualization is a method of moving information storage from physical servers to virtual ones, often in a different location. In the past, large companies would keep physical servers on site that held huge amounts of corporate information. These servers were expensive, both to purchase and maintain. With data center virtualization, it became possible to separate both the hardware and location from the data. This cuts costs and increases the data's availability. Data center virtualization actually comes from a combination of two different technologies; high-speed data transfer and server virtualization. Without both of these components, data center virtualization becomes highly impractical.

There are three areas of IT where virtualization is making inroads, **network virtualization, storage virtualization and server virtualization.**

Network virtualization is a method of combining the available resources in a network by splitting up the available bandwidth into channels, each of which is independent from the others, and each of which can be assigned (or reassigned) to a particular server or device in real time. The idea is that virtualization disguises the true complexity of the network by separating it into manageable parts, much like your partitioned hard drive makes it easier to manage your files.

Storage virtualization is the pooling of physical storage from multiple network storage devices into what appears to be a single storage device that is managed from a central console. Storage virtualization is commonly used in storage area networks (SANs).

Server virtualization is the masking of server resources, including the number and identity of individual physical servers, processors, and operating systems, from server users. The server administrator uses a software

application to divide one physical server into multiple isolated virtual environments. The virtual environments are sometimes called virtual private servers, but they are also known as guests, instances, containers or emulations.

Benefits of Data Center Virtualization

Following are the benefits in addition to the cost savings that result from reducing the number of servers:

- Enables the consolidation of physical servers, slashing the costs of operating a data center. This includes reducing the costs of server upgrades, management, power, space, and storage.
- Reduction in data center space and in data center equipment such as PDUs, air conditioning units, etc.
- Reduction in the number of network, HBAs and SAN switches.
- Provides true high-availability for all servers without requiring duplicate hardware and clustering software.
- Integrates the test/development and production environments while significantly enhancing the test/development process.
- Facilitates true disaster recovery for all servers.
- Eliminates the need for maintenance windows for physical server troubleshooting or upgrades and enables faster server provisioning.
- Enhances security and provides regulatory compliance benefits.

Cloud Data Center

A cloud data center has three distinct characteristics that differentiate it from traditional Data Center. It is sold on demand, typically by the minute or the hour; it is elastic -- a user can have as much or as little of a service as they want at any given time; and the service is fully managed by the provider (the consumer needs nothing but a personal computer and Internet access). Significant innovations in virtualization and distributed computing, as well as improved access to high-speed Internet and a weak economy, have accelerated interest in cloud computing.

The multiplier effect of Internet data, trends of enterprises information transformation and tremendous load bring forth traditional Data Center huge challenges: how to reduce operation and maintenance cost, how to meet demand of high capacity, high security and high efficiency. Attributing to resource on-demand, flexible and dynamic structure, cloud computing is the right technology to resolve all these issues. By continuously improving core technologies including virtualization, elastic computing and high-density computing etc, cloud vendors have creatively developed Modular cloud computing Data Center of trusty, efficient, smart, ultra-bandwidth and green end to end solution. Cloud Data Center is now evolving beyond being merely a model of technology delivery to becoming a new operating model where business decision makers are empowered to procure infrastructure on demand, and where IT becomes an internal service provider delivering increased business agility without compromising security or control.

Traditional Corporate Data Center	**Cloud Data Center**
Thousands of different applications	Few applications (maybe even just one)
Mixed hardware environment	Homogeneous hardware environment
Multiple management tools	Standardized management tools
Frequent application patching and updating	Minimal application patching and updating
Complex workloads	Simple workloads
Multiple software architectures	Single standard software architecture

Service Oriented Architecture

If a service-oriented architecture is to be effective, we need a clear understanding of the term service.

A service is a function that is well-defined, self-contained, and does not depend on the context or state of other services. A service-oriented architecture is essentially a collection of services. These services communicate with each other. The communication can involve either simple data passing or it could involve two or more services coordinating some activity.

SOA defines how to integrate widely disparate applications for a Web-based environment and uses multiple implementation platforms. Rather than defining an API, SOA defines the interface in terms of protocols and functionality.

Service-oriented architectures are not new. The first service-oriented architectures are usually considered to be the Distributed Component Object Model (DCOM) or Object Request Brokers (ORBs), which were based on the Common Object Requesting Broker Architecture (CORBA) specification. The introduction of SOA provides a platform for technology and business units to meet business requirements of the modern enterprise. With SOA, our organization can use existing application systems to a greater extent and may respond faster to change requests.

Why SOA?

The reality in IT enterprises is that infrastructure is heterogeneous across operating systems, applications, system software, and application infrastructure. Some existing applications are used to run current business processes, so starting from scratch to build new infrastructure isn't an option. nterprises should quickly respond to business changes with agility; leverage existing investments in applications and application infrastructure to address newer business requirements; support new channels of interactions with customers, partners, and suppliers; and feature an architecture that supports organic business.

SOA with its loosely coupled nature allows enterprises to plug in new services or upgrade existing services in a granular fashion to address the new business requirements, provides the option to make the services consumable across different channels, and exposes the existing enterprise and legacy applications as services, thereby safeguarding existing IT infrastructure investments.

For example, a core banking application provides a Fund Transfer service, then the other banking applications such as Treasury, Payment Gateway, ATM Switching, and so on can call or invoke Fund Transfer service without need to worry about where the Fund Transfer is located in the network. This contrasts with the Tight Coupling approach. Each application defining their own Fund Transfer, the problem come when the Transfer Fund logic is change. It will be difficult and require high cost (and time, of course) to set the new logic into each application.

Component Object Model and Object Request Broker Architectures

DCOM (Distributed Component Object Model) is a set of Microsoft concepts and program interfaces in which client program objects can request services from server program objects on other computers in a network. DCOM is based on the Component Object Model (COM), which provides a set of interfaces allowing clients and servers to communicate within the same computer (that is running Windows 95 or a later version).

Common Object Request Broker Architecture (CORBA) is an architecture and specification for creating, distributing, and managing distributed program objects in a network. In Common Object Request Broker Architecture (CORBA), an Object Request Broker (ORB) is the programming that acts as a "broker" between a client request for a service from a distributed object or component and the completion of that request.

Characteristics of SOA

- In SOA, Services should be independent of other services. Altering a service should not affect calling service.
- Services should be self-contained. When we talk about a Register Customer service it means, service will do all the necessary work for us, we are not required to care about anything.
- Services should be able to define themselves. Services should be able to answer a question what is does? It should be able to tell client what all operations it does, what all data types it uses and what kind of responses it will return.
- Services should be published into a location (directory) where anyone can search for it.
- As said, SOA comprises of collection services which communicate via standard Messages. Standard messages make them platform independent. (Here standard doesn't mean standard across Microsoft it means across all programming languages and technologies.)
- Services should be able to communicate with each other asynchronously.
- Services should support reliable messaging. Means there should be a guarantee that request will be reached to correct destination and correct response will be obtained.
- Services should support secure communication.

Open source software

Open-source software (OSS) is computer software with its source code made available and licensed with an open-source license in which the copyright holder provides the rights to study, change and distribute the software for free to anyone and for any purpose. Open-source software is very often developed in a public, collaborative manner. The basic behind the Open Source Initiative is that when programmers can read, redistribute and modify the source code for a piece of software, the software evolves. Open source sprouted in the technological community as a response to proprietary software owned by corporations.

Proprietary software is privately owned and controlled. In the computer industry, proprietary is considered the opposite of open. A proprietary design or technique is one that is owned by a company. It also implies that the company has not divulged specifications that would allow other companies to duplicate the product.

Open Source is a certification standard issued by the Open Source Initiative (OSI) that indicates that the source code of a computer program is made available free of charge to the general public. OSI dictates that in order to be considered "OSI Certified" a product must meet the following criteria:

- The author or holder of the license of the source code cannot collect royalties on the distribution of the program.
- The distributed program must make the source code accessible to the user.
- The author must allow modifications and derivations of the work under the program's original name.
- No person, group or field of endeavor can be denied access to the program.

- The rights attached to the program must not depend on the program's being part of a particular software distribution.
- The licensed software cannot place restrictions on other software that is distributed with it.

MySQL

MySQL is the preferred open source database based on usage. According to the MySQL web site, 15 it has become the world's most popular open source database. It is used by individual developers as well as by many of the world's largest companies, including Yahoo!, Alcatel-Lucent, Google, Nokia, YouTube, and Zappos.com. MySQL runs on more than 20 platforms, including Linux, Windows, OS/X, HP-UX, AIX, and Netware. Users can freely download and deploy MySQL from the official web site without cost. This product is in use in millions of small to medium-scale applications. MySQL is the preferred database in LAMP architecture (Linux/Apache/MySQL/PHP-Python-Perl). This regal position affords MySQL access to over two-thirds of the world's web database servers. MySQL is deployed with nearly every Linux distribution, and is easily installed on Windows, Mac, and Solaris platforms for both server and client use. In the cloud, MySQL is the king of the database server packages because it is proven, reliable, scalable, and free.

Until the release of version 5.1, MySQL had to take a back seat to commercial enterprise-grade database products such as Oracle and IBM's DB2 because of a lack of clustering, partitioning, and replication features. With the 5.1 release, those hurdles were overcome. Later, spatial data, JSON, and native XML support are also embedded.

PostgreSQL

PostgreSQL is another powerful open source DBMS. According to the official web site, 16 it has more than 15 years of active development and a proven architecture that has earned it a strong reputation for reliability, data integrity, and correctness. It runs on all major operating systems and prides itself in standards compliance. PostgreSQL has a fully relational system catalog which itself supports multiple schemas per database.

PostgreSQL is highly scalable, both in the magnitude of data it can manage and in the number of concurrent users it can accommodate. There are active PostgreSQL systems in production environments that manage in excess of 4 TB of data. For larger cloud implementations, PostgreSQL may be the DBMS of choice. Another important point to consider for any cloud implementation of a database tier is the security of the database. Accordingly, PostgreSQL is considered by many to be the most secure out-of-the-box configuration available for a database. PostgreSQL boasts many sophisticated features and is another good choice for cloud computing applications.

Cloud Computing Security

Cloud computing security is the set of control-based technologies and policies designed to adhere to regulatory compliance rules and protect information, data applications and infrastructure associated with cloud computing use. Cloud Security Service is a service provider solution that provides cloud hosted offerings. It allows customers to federate to their web-based and SaaS applications through their provider, and lets service providers connect to their customers through a secure channel without requiring them to open their firewall ports.

Cloud Security Issues and Challenges

Heightened security threats must be overcome in order to benefit fully from this new computing paradigm. Some security concerns are listed and discussed below:

- Security concern #1: With the cloud model control physical security is lost because of sharing computing resources with other companies. No knowledge or control of where the resources run.
- Security concern #2: Company has violated the law(risk of data seizure by (foreign) government).
- Security concern #3: Storage services provided by one cloud vendor may be incompatible with another vendor's services if user decides to move from one to the other (e.g. Microsoft cloud is incompatible with Google cloud).
- Security concern #4: Who controls the encryption/decryption keys? Logically it should be the customer.
- Security concern #5: Ensuring the integrity of the data(transfer, storage, and retrieval) really means that it changes only in response to authorized transactions. A common standard to ensure data integrity does not yet exist.
- Security concern #6: In case of Payment Card Industry Data Security Standard (PCI DSS) data logs must be provided to security managers and regulators.
- Security concern #7: Users must keep up to date with application improvements to be sure they are protected.
- Security concern #8: Some government regulations have strict limits on what data about its citizens can be stored and for how long, and some banking regulators require that customer's financial data remain in their home country.
- Security concern #9: The dynamic and fluid nature of virtual machines will make it difficult to maintain the consistency of security and ensure the audit ability of records.
- Security concern #10: Customers may be able to sue cloud service providers if their privacy rights are violated, and in any case the cloud service providers may face damage to their reputation. Concerns arise when it is not clear to individuals why their personal information is requested or how it will be used or passed on to other parties.

Security Issues

- Access: Data subjects have a right to know what personal information is held and, in some cases, can make a request to stop processing it. If a data subject exercises this right to ask the organization to delete his data, will it be possible to ensure that all of his information has been deleted in the cloud?
- Compliance: What are the applicable laws, regulations, standards, and contractual commitments that govern this information, and who is responsible for maintaining the compliance? Clouds can cross multiple jurisdictions in multiple states.

- Storage: Where is the data in the cloud stored? Was it transferred to another data center in another country? Privacy laws in various countries place limitations on the ability of organizations to transfer some types of personal information to other countries.
- Retention: How long is personal information (that is transferred to the cloud) retained? Who enforces the retention policy in the cloud, and how are exceptions to this policy (such as litigation holds) managed?
- Destruction: How can we know that the cloud service provider (CSP) didn't retain additional copies? Did the CSP really destroy the data, or just make it inaccessible to the organization? Is the CSP keeping the information longer than necessary so that it can mine the data for its own use?
- Audit and monitoring: How can organizations monitor their CSP and provide assurance to relevant stakeholders that privacy requirements are met when their PII is in the cloud?
- Privacy breaches: How can we ensure that the cloud service provider (CSP) notifies us when a breach occurs, and who is responsible for managing the breach notification process (and costs associated with the process)? If contracts include liability for breaches resulting from negligence of the CSP, how is the contract enforced and how is it determined who is at fault?

Securing Data in the Cloud

A secure infrastructure ensures and builds confidence that the data stored is secure in providers' side. Proper implementation of security measures is mandatory in cloud computing. The fact that application is launched over the internet makes it susceptible for security risks. Physical location of stored data is also vital and it's the responsibility of the provider to choose the right location of storage. Restricted user access can vary from simple user name / password protection to CAPTCHA log in forms. When an employee no longer has a business need to access data center his privileges to access datacenter should be immediately revoked. Cloud Providers can also consider one time password authentication where the clients will get one time temporary password from SSN /mobile device which helps in data security even if password is compromised.

The following practices can be followed to improvise the security of cloud computing:

- Implement security practices at organizational level and make sure that the providers security plans are in alignment with the business.
- Employ and maintain secure Infrastructure in client side (secure VPN , changing default vendor provided passwords) and host side (firewalls , patch managements ,anti-virus updates)
- Ensure accurate user permissions and restricted access at both sides
- Data sanitizations at the right time
- Include clause regarding data ownership and protection of intellectual property in Service Level Agreement (SLA).
- Keep a log of Users who access data, time of event and event description.
- Regular auditing should be conducted.
- Encryption /decryption key should be kept secured.
- Providers should verify the authenticity of their clients.
- Frequent data backup policy should be in place
- Penetration testing at regular intervals to ensure vulnerabilities is not in the cloud.

Software-as-a-Service (SaaS) Security Issues

Cloud computing models of the future will likely combine the use of SaaS (and other as a service as appropriate), utility computing, and Web 2.0 collaboration technologies to leverage the Internet to satisfy their customers' needs. New business models being developed as a result of the move to cloud computing are creating not only new technologies and business operational processes but also new security requirements and challenges as described previously.

SaaS will likely remain the dominant cloud service model for the foreseeable future and the area where the most critical need for security practices and oversight will reside. Just as with an managed service provider, corporations or end users will need to research vendors' policies on data security before using vendor services to avoid losing or not being able to access their data.

In Software as a Service (SaaS) model, the client needs to be dependent on the service provider for proper security measures of the system. The service provider must ensure that their multiple users don't get to see each other's private data. So, it becomes important to the user to ensure that right security measures are in place and also difficult to get an assurance that the application will be available when needed. Cloud computing providers need to provide some solution to solve the common security challenges that traditional communication systems face. At the same time, they also have to deal with other issues inherently introduced by the cloud computing paradigm itself.

- Authentication and authorization
  The authorization and authentication applications used in enterprise environments need to be changed, so that they can work with a safe cloud environment. Forensics tasks will become much more difficult since it will be very hard or maybe not possible for investigators may to access the system hardware physically.
- Data confidentiality
  Confidentiality may refer to the prevention of unintentional or intentional unauthorized disclosure or distribution of secured private information. Confidentiality is closely related to the areas of encryption, intellectual property rights, traffic analysis, covert channels, and inference in cloud system. Whenever a business, an individual, a government agency, or any other entity wants to shares information over cloud, confidentiality or privacy is a questions may need to be asked.
- Availability
  The availability ensures the reliable and timely access to cloud data or cloud computing resources by the appropriate personnel. The availability is one of the big concerns of cloud service providers, since if the cloud service is disrupted or compromised in any way; it affects large no. of customers than in the traditional model.
- Information Security
  In the SaaS model, the data of enterprise is stored outside of the enterprise boundary, which is at the SaaS vendor premises. Consequently, these SaaS vendor needs to adopt additional security features to ensure data security and prevent breaches due to security vulnerabilities in the application or by malicious employees. This will need the use of very strong encryption techniques for data security and highly competent authorization to control access private data.
- Data Access

Data access issue is mainly related to security policies provided to the users while accessing the data. Organizations have their own security policies based on which each employee can have access to a particular set of data. The SaaS model must be flexible enough to incorporate the specific policies put forward by the organization.

- Network Security
  In a SaaS deployment model, highly sensitive information is obtained from the various enterprises, then processed by the SaaS application and stored at the SaaS vendor's premises. All data flow over the network has to be secured in order to prevent leakage of sensitive information.
- Data breaches
  Since data from various users and business organizations lie together in a cloud environment, breaching into this environment will potentially make the data of all the users vulnerable. Thus, the cloud becomes a high potential target.
- Identity management and sign-on process
  Identity management (IdM) or ID management is an area that deals with identifying individuals in a system and controlling the access to the resources in that system by placing restrictions on the established identities. Area of IdM is considered as one of the biggest challenges in information security. When a SaaS provider want to know how to control who has access to what systems within the enterprise it becomes a lot more challenging task.

Security Management

Lack of clearly defined roles and responsibilities, and agreement on expectations, can result in a general feeling of loss and confusion among the security team about what is expected of them, how their skills and experienced can be leveraged, and meeting their performance goals. Morale among the team and pride in the team is lowered, and security suffers as a result.

Risk Management

Effective risk management entails identification of technology assets; identification of data and its links to business processes, applications, and data stores; and assignment of ownership and protective responsibilities. Actions should also include maintaining a repository of information assets. Owners have authority and accountability for information assets including protection requirements, and protection implement confidentiality, integrity, availability, and privacy controls. A formal risk assessment process should be created that allocates security resources linked to business continuity.

Risk/ Vulnerability Assessment

Security risk assessment is critical to helping the information security organization make informed decisions when balancing the conflicting priorities of business utility and protection of assets. Lack of attention to completing formalized risk assessments can contribute to an increase in information security audit findings, can make vulnerable certification goals, and can lead to inefficient and ineffective selection of security controls that may not adequately mitigate information security risks to an acceptable level. A formal information security risk management process should proactively assess information security risks as well as plan and manage them on a periodic or as-needed basis.

Security Monitoring and Incident Response

Centralized security information management systems should be used to provide notification of security vulnerabilities and to monitor systems continuously through automated technologies to identify potential issues. They should be integrated with network and other systems monitoring processes (e.g., security information management, security event management, security information and event management, and security operations centers that use these systems for dedicated 24/7/365 monitoring). Management of periodic, independent third-party security testing should also be included.

Many of the security threats and issues in SaaS center around application and data layers. So the types and sophistication of threats and attacks for a SaaS organization require a different approach to security monitoring than traditional infrastructure and perimeter monitoring. The organization may thus need to expand its security monitoring capabilities to include application- and data-level activities. This may also require subject-matter experts in applications security and the unique aspects of maintaining privacy in the cloud. Without this capability and expertise, a company may be unable to detect and prevent security threat and attacks to its customer data and service stability.

Incident response is an organized approach to addressing and managing the outcome of a security breach or attack (also known as an incident). The goal is to handle the situation in a way that limits damage and reduces recovery time and costs. An incident response plan includes a policy that defines, in specific terms, what constitutes an incident and provides a step-by-step process that should be followed when an incident occurs. An organization's incident response is conducted by the computer incident response team, a carefully selected group that, in addition to security and general IT staff, may include representatives from legal, human resources, and public relations departments.

Security Architecture Design

Security Architecture is one component of a products/systems overall architecture and is developed to provide guidance during the design of the product/system. A security architecture framework should be established with consideration of processes (enterprise authentication and authorization, access control, confidentiality, integrity, non- repudiation, security management, etc.), operational procedures, technology specifications, people and organizational management, and security program compliance and reporting.

A security architecture document should be developed that defines security and privacy principles to meet business objectives. Documentation is required for management controls and metrics specific to asset classification and control, physical security, system access controls, network and computer management, application development and maintenance, business continuity, and compliance. The creation of a secure architecture provides the engineers, data center operations personnel, and network operations personnel a common blueprint to design, build, and test the security of the applications and systems. Design reviews of new changes can be better assessed against this architecture to assure that they conform to the principles described in the architecture, allowing for more consistent and effective design reviews.

Vulnerability Assessment

Vulnerability assessment classifies network assets to more efficiently prioritize vulnerability- mitigation programs, such as patching and system upgrading. It measures the effectiveness of risk mitigation by setting goals of reduced vulnerability exposure and faster mitigation. Vulnerability management should be integrated with discovery, patch

management, and upgrade management processes to close vulnerabilities before they can be exploited. A vulnerability assessment attempts to identify the exposed vulnerabilities of a specific host, or possibly an entire network. The vulnerabilities may be due to configuration problems or missing software patches. Vulnerability Assessment in cloud should be done in periodic basis with predefined service level agreement. Customers should be allowed to test cloud infrastructure before and after they outsource their infrastructure to cloud.

Data Privacy and Security

Cloud computing has transformed the way organizations approach IT, enabling them to become more agile, introduce new business models, provide more services, and reduce IT costs. Cloud computing technologies can be implemented in a wide variety of architectures, under different service and deployment models, and can coexist with other technologies and software design approaches. Maintaining control over the data is paramount to cloud success. A decade ago, enterprise data typically resided in the organization's physical infrastructure, on its own servers in the enterprise's data center, where one could segregate sensitive data in individual physical servers. Today, with virtualization and the cloud, data may be under the organization's logical control, but physically reside in infrastructure owned and managed by another entity.

This shift in control is the number one reason new approaches and techniques are required to ensure organizations can maintain data security. When an outside party owns, controls, and manages infrastructure and computational resources, how can you be assured that business or regulatory data remains private and secure, and that your organization is protected from damaging data breaches—and feel you can still completely satisfy the full range of reporting, compliance, and regulatory requirements? Some of the points to keep data private and secure in cloud infrastructure are as below:

- Avoid storing sensitive information in the cloud.
- Read the user agreement to find out how your cloud service storage works.
- Password sensitivity
- Encrypt your data
- Use Encrypted cloud services

Application Security

External penetration testers are used for application source code reviews, and attack and penetration tests provide an objective review of the security of the application as well as assurance to customers that attack and penetration tests are performed regularly. Fragmented and undefined collaboration on application security can result in lower-quality design, coding efforts, and testing results. Some of the things that we should consider while moving to cloud application are:

- Risks associated with cloud application
- The fact that someone is managing and controlling your critical application
- The perimeter of cloud is different and multitenant
- Application should be protected with industry standard firewall and security products
- Insecure Interfaces and Application Program Interface (API's)
- Denial of Service (DOS) attack

Virtual Machine Security

Virtual machines are the containers in which applications and guest operating systems run. By design, all VMware virtual machines are isolated from one another. This isolation enables multiple virtual machines to run securely while sharing hardware and ensures both their ability to access hardware and their uninterrupted performance. Each virtual machine is isolated from other virtual machines running on the same hardware. Although virtual machines share physical resources such as CPU, memory, and I/O devices, a guest operating system on an individual virtual machine cannot detect any device other than the virtual devices made available to it.

In the cloud environment, physical servers are consolidated to multiple virtual machine instances on virtualized servers. Not only can data center security teams replicate typical security controls for the data center at large to secure the virtual machines, they can also advise their customers on how to prepare these machines for migration to a cloud environment when appropriate. Firewalls, intrusion detection and prevention, integrity monitoring, and log inspection can all be deployed as software on virtual machines to increase protection and maintain compliance integrity of servers and applications as virtual resources move from on-premises to public cloud environments. By deploying this traditional line of defense to the virtual machine itself, we can enable critical applications and data to be moved to the cloud securely.

To facilitate the centralized management of a server firewall policy, the security software loaded onto a virtual machine should include a bidirectional stateful firewall that enables virtual machine isolation and location awareness, thereby enabling a tightened policy and the flexibility to move the virtual machine from on-premises to cloud resources. Integrity monitoring and log inspection software must be applied at the virtual machine level. A further area of concern with virtualization has to do with the potential for undetected network attacks between VMs collocated on a physical server. Unless we can monitor the traffic from each VM, we can't verify that traffic isn't possible between those VMs. In essence, network virtualization must deliver an appropriate network interface to the VM. That interface might be a multiplexed channel with all the switching and routing handled in the network interconnect hardware.

Disaster Recovery

A Disaster Recovery Plan (DRP) is a business plan that describes how work can be resumed quickly and effectively after a disaster. Disaster recovery planning is just part of business continuity planning and applied to aspects of an organization that rely on an IT infrastructure to function. The overall idea is to develop a plan that will allow the IT department to recover enough data and system functionality to allow a business or organization to operate - even possibly at a minimal level. A disaster recovery plan (DRP) documents policies, procedures and actions to limit the disruption to an organization in the wake of a disaster. Just as a disaster is an event that makes the continuation of normal functions impossible, a disaster recovery plan consists of actions intended to minimize the negative effects of a disaster and allow the organization to maintain or quickly resume mission-critical functions. To better understand and evaluate disaster recovery strategies, it is important to define two terms: recovery time objective (RTO) and recovery point objective (RPO).

The recovery time objective (RTO) is the maximum amount of time allocated for restoring application functionality. This is based on business requirements and is related to the importance of the application. Critical business applications require a low RTO. The recovery point objective (RPO) is the acceptable time window of lost data due to the recovery process. For example, if the RPO is one hour, you must completely back up or replicate

the data at least every hour. Once you bring up the application in an alternate datacenter, the backup data may be missing up to an hour of data. Like RTO, critical applications target a much smaller RPO.

Why Disaster Recovery is needed?

- Machines, hardware and even data centers fail.
- Much like machines, humans are not perfect. They make mistakes. In case of mistakes, DR may help resume business from back date.
- Customers expect perfection as they don't want disruption in services
- DR enabled organizations will attract more customers.

Disaster Recovery Management/ Planning Steps

- Count the costs
  Although data center downtime is harmful to any company that relies on its IT services, it costs some companies more than others. Our disaster recovery plan should enable a fast return to service, but it shouldn't cost us more than we are losing in downtime costs.
- Evaluate the types of threats you face and how extensively they can affect your facility
  Malicious attacks can occur anywhere, but we may also face threats peculiar to our location, such as weather events (tornadoes, hurricanes, floods and so on), earthquakes or other dangers. Part of preparing for a disaster is to know what is likely to occur and how those threats could affect our systems. Evaluating these situations beforehand allows us to better take appropriate action should one of these events occur.
- Know what you have and how critical it is to operations
  Responding to a disaster in your data center is similar to doing so in medicine: you need to treat the more serious problems first, then the more minor ones. By determining which systems are most critical to your data center, you enable your IT staff to prioritize and make the best use of the precious minutes and hours immediately following an outage. Not every system need be functional immediately following a disaster.
- Identify critical personnel and gather their contact information
  Who do you most want to be present in the data center following an outage? Who has the most expertise in a given area and the greatest ability to oversee some part of the recovery effort? Being able to get in touch with these people is crucial to a fast recovery. Collect their contact information and, just as importantly, keep it up to date. If it's been a year or more since you last checked, some of that contact information is likely out of date. Every minute you spend trying to find important personnel is time not spent on recovery.
- Train your employees
  Knowledge of how to implement disaster recovery procedures is obviously important when an outage occurs. To this end, prepare by training personnel— and not just in their respective areas of expertise. Everyone should have some broad- based knowledge of the recovery process so that it can be at least started even if not everyone is present.
- Ensure that everyone knows the disaster recovery plan and understands his or her role
  Announcing the plan and assigning roles is not something you should do after a disaster strikes; it should be done well in advance, leaving time for personnel to learn their roles and to practice them. Almost nothing about a disaster event should be new (aside from some contingencies of the moment, perhaps): the IT staff should implement disaster recovery as a periodic task (almost) like any other.

- Practice
  Needless to say, this is perhaps the most critical part of preparation for a downtime event. The difference between knowing your role and being able to execute it well is simply practice. We may not be able to shut down your data center to simulate precisely all of the conditions we will face in an outage, but we can go through many of the procedures nevertheless.
- Automate where possible
  Our staff is limited, so it can only do so much. The more that our systems can do on their own in a recovery situation, the faster the recovery will generally be. This also leaves less room for human error—particularly in the kind of stressful atmosphere that exists following a disaster.
- Follow up after a disaster
  When a downtime event does occur, evaluate the performance of the personnel and the plan to determine if any improvements can be made. Update your plan accordingly to enable a better response in the future. Furthermore, investigate the cause of the outage. If it's an internal problem, take necessary measures to correct equipment issues to avoid the same problem occurring again.